

MT2 Exam solution

Problem 1

- a. Determine, with or without calculations, the force exerted by the outer ring on the electron.

Due to symmetry, the force exerted by the outer ring on the e^- is 0.

$$\vec{F}_{\text{out}} = 0$$

- b. Determine the force exerted by the truncated ring on the electron.

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dQ}{R^2} \hat{r}$$

$$dQ = \lambda R d\theta = \frac{Q}{2\pi} d\theta \quad x = R \cos\theta$$

use polar coordinate (R, θ) , where $y = R \sin\theta$
on the truncated ring, $R = R$, $\hat{r} = (\cos\theta, \sin\theta)$

$$\vec{E} = \int d\vec{E} = \frac{1}{4\pi\epsilon_0} \int_0^\pi \frac{Q}{\pi R^2} d\theta (\cos\theta \hat{x} + \sin\theta \hat{y})$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{\pi R^2} \times 2 \hat{y} = \frac{Q}{2\pi^2\epsilon_0 R^2} \hat{y}$$

$$\vec{F}_{\text{inj}} = (-e) \vec{E} = \frac{eQ}{2\pi^2\epsilon_0 R^2} (-\hat{y}) \quad \text{direction is along } (-\hat{y})$$

c. Determine the force exerted by the two point-charges on the electron.

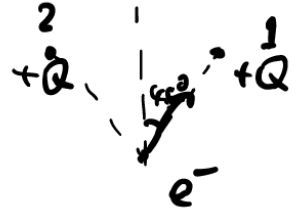
$$\vec{F}_1 = \frac{1}{4\pi\epsilon_0} \frac{eQ}{R^2} \hat{r}_1$$

$$\hat{r}_1 = (\cos 45^\circ, \sin 45^\circ) = \frac{\sqrt{2}}{2} \hat{x} + \frac{\sqrt{2}}{2} \hat{y}$$

$$\vec{F}_2 = \frac{1}{4\pi\epsilon_0} \frac{eQ}{R^2} \hat{r}_2$$

$$\hat{r}_2 = -\frac{\sqrt{2}}{2} \hat{x} + \frac{\sqrt{2}}{2} \hat{y}$$

$$\begin{aligned} \vec{F}_{\text{point}} &= \vec{F}_1 + \vec{F}_2 = \frac{1}{4\pi\epsilon_0} \frac{eQ}{R^2} \cdot \sqrt{2} \hat{y} \\ &= \frac{eQ}{2\sqrt{2}\pi\epsilon_0 R^2} \hat{y} \end{aligned}$$



d. Could the electron - of mass m - remain at rest at the origin if the smiley face lies in a vertical plane? If so, determine charge Q . Note that $\pi > \sqrt{2}$.

$$\vec{F}_{\text{total}} = \vec{F}_{\text{act}} + \vec{F}_{\text{ring}} + \vec{F}_{\text{point}}$$

$$= \frac{eQ}{2\pi^2\epsilon_0 R^2} (-\hat{y}) + \frac{eQ}{2\sqrt{2}\pi\epsilon_0 R^2} \hat{y}$$

$$= \frac{eQ}{2\pi\epsilon_0 R^2} \left(\frac{1}{\sqrt{2}} - \frac{1}{\pi} \right) \hat{y}$$

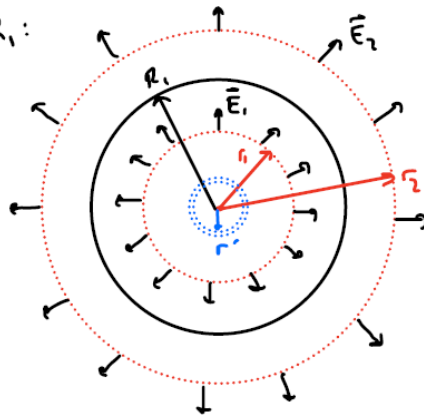
since $\pi > \sqrt{2}$ $\frac{1}{\sqrt{2}} - \frac{1}{\pi} > 0$ it can rest

$$\frac{eQ}{2\pi\epsilon_0 R^2} \left(\frac{1}{\sqrt{2}} - \frac{1}{\pi} \right) = mg \Rightarrow Q = \frac{2\pi\epsilon_0 R^2 mg}{e \left(\frac{1}{\sqrt{2}} - \frac{1}{\pi} \right)}$$

Problem 2

a. Determine the magnitude and direction of the electric field produced by the charged sphere at any point, both inside and outside the sphere. Explain your choice of method to calculate the electric field.

The charge is arranged in a spherically symmetric fashion, with ρ being a function of r only, so the field will be spherically symmetric, always pointing parallel to $\hat{r}$ with magnitude dependent only on r . So, we'll use Gauss's law with spherical Gaussian surfaces concentric with the charged sphere. To find the field inside the sphere, we'll use a Gaussian surface of radius $r_1 < R_1$, and to find the field outside the sphere we'll use a Gaussian surface of radius $r_2 > R_1$:



As shown, the fields $\vec{E}_1$ at r_1 and $\vec{E}_2$ at r_2 always cross the Gaussian surfaces perpendicularly through them, parallel to the area vectors, and at constant magnitude in each, so

$$\begin{aligned}\Phi_1 &= \oint \vec{E}_1 \cdot d\vec{A}_1 = \oint E_1 dA_1 = E_1 \oint dA_1 \\ &= E_1 A_1 = E_1 \cdot 4\pi r_1^2\end{aligned}$$

and similarly $\Phi_2 = E_2 \cdot 4\pi r_2^2$. To find charge enclosed, we find the charge of a thin spherical shell at radius r' with thickness dr' inside the Gaussian surface and integrate over these shells.

Such a shell has surface area $A = 4\pi r'^2$ and thickness dr' , so the volume of a shell is $dV = A dr' = 4\pi r'^2 dr'$. The charge is then $dQ = \rho(r') dV = \frac{\rho_0 R_1}{r'} \cdot 4\pi r'^2 dr' = 4\pi \rho_0 R_1 r' dr'$,

and so

$$Q_1 = \int dQ = \int_0^{r_1} 4\pi \rho_0 R_1 r' dr' = 2\pi \rho_0 R_1 r'^2 \Big|_{r=0}^{r=r_1} = 2\pi \rho_0 R_1 r_1^2$$

$$Q_2 = \int dQ = \int_0^{R_1} 4\pi \rho_0 R_1 r' dr' = 2\pi \rho_0 R_1 r'^2 \Big|_{r=0}^{r=R_1} = 2\pi \rho_0 R_1^3 \quad (\text{only integrate to } R_1 \text{ since } R_1 \text{ to } r_2 \text{ is empty of charge})$$

Gauss's law says that

$$\Phi_1 = \frac{Q_1}{\epsilon_0} \Rightarrow E_1 \cdot 4\pi r_1^2 = \frac{2\pi \rho_0 R_1 r_1^2}{\epsilon_0} \Rightarrow E_1 = \frac{\rho_0 R_1}{2\epsilon_0}$$

$$\Phi_2 = \frac{Q_2}{\epsilon_0} \Rightarrow E_2 \cdot 4\pi r_2^2 = \frac{2\pi \rho_0 R_1^3}{\epsilon_0} \Rightarrow E_2 = \frac{\rho_0 R_1^3}{2\epsilon_0 r_2^2}$$

$$\vec{E} = \frac{\rho_0 R_1}{2\epsilon_0} \begin{cases} \hat{r}, & r < R_1 \\ \frac{R_1^3}{r^2} \hat{r}, & r > R_1 \end{cases}$$

b. Determine the electric potential at any point ($r \leq R_1$ and $r \geq R_1$), assuming that $V(0)=0$.

We know that $\Delta V(\mathbf{r}) = - \int_0^{\mathbf{r}} \vec{E} \cdot d\vec{r}$, so to find the potential $V(r)$ we'll use

$$V(r) = V(0) + (V(r) - V(0)) = V(0) + \Delta V(r: 0 \rightarrow r: r) = - \int_0^r \vec{E}(r') \cdot d\vec{r}' \\ = - \int_0^r E(r') dr',$$

where we use that $\vec{E}(r') = E(r')\hat{r}$ and $d\vec{r}' = dr'\hat{r}$ for a path radially outward in getting to the last line. For $r < R_1$, we get that

$$V(r) = - \int_0^r \frac{\rho_0 R_1}{2\epsilon_0} dr' = - \frac{\rho_0 R_1}{2\epsilon_0} r' \Big|_{r'=0}^{r'=r} = - \frac{\rho_0 R_1}{2\epsilon_0} r,$$

and since V will be continuous $V(R_1) = - \frac{\rho_0 R_1}{2\epsilon_0} \cdot R_1 = - \frac{\rho_0 R_1^2}{2\epsilon_0}$. Then, to find $V(r)$ for $r > R_1$ we can use

$$V(r) = V(R_1) + \Delta V(r: R_1 \rightarrow r: r) = - \frac{\rho_0 R_1^2}{2\epsilon_0} - \int_{R_1}^r \frac{\rho_0 R_1^3}{2\epsilon_0 r'^2} dr' = - \frac{\rho_0 R_1^2}{2\epsilon_0} + \frac{\rho_0 R_1^3}{2\epsilon_0 r'} \Big|_{r'=R_1}^{r'=r} \\ = - \frac{\rho_0 R_1^2}{2\epsilon_0} + \frac{\rho_0 R_1^3}{2\epsilon_0 r} - \frac{\rho_0 R_1^2}{2\epsilon_0} = \frac{\rho_0 R_1^2}{2\epsilon_0} \left(\frac{R_1}{r} - 2 \right)$$

so, in total

$$V(r) = \frac{\rho_0 R_1^2}{2\epsilon_0} \cdot \begin{cases} -\frac{r}{R_1}, & r < R_1 \\ \frac{R_1}{r} - 2, & r > R_1 \end{cases}$$

A conducting spherical shell extending from R_1 to R_2 [gray shade] (Fig.2) is added around the previous non-conducting sphere [hatching lines] and the electric field is measured to be zero outside of the combined system ($r > R_2$).

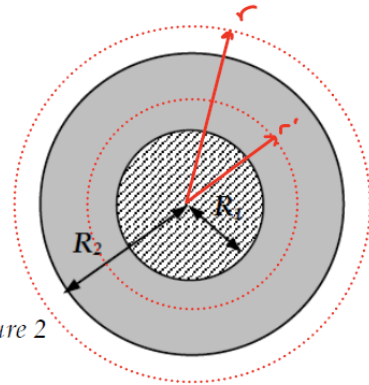


Figure 2

c. Determine the net electric charge Q_c carried by the conducting sphere.

We draw a Gaussian sphere of radius $r > R_2$ outside the conductor. Like in a, $\Phi = E \cdot 4\pi r^2$, but $E = 0$ so $\Phi = 0$. By Gauss's law, $E = \frac{Q_{enc}}{\epsilon_0}$ so $Q_{enc} = 0$ as well. The charge of the non-conducting sphere is what we calculated earlier as $Q_1 = 2\pi\rho_0 R_1^3$, so to cancel this we need that

$$Q_c = -Q_1 = -2\pi\rho_0 R_1^3$$

d. Determine the surface and volume charge densities for the conducting spherical shell. Let σ_1 be the surface charge density on the inner surface, σ_2 the surface charge density on the outer surface, and ρ_c the volumetric charge density inside the conducting spherical shell. Make sure you articulate your reasoning very clearly.

Looking at a second Gaussian surface at radius r' between R_1 and R_2 , again

$\Phi = E \cdot 4\pi r'^2$, but $E = 0$ since r' is inside the conducting material, and therefore

we need that $Q_{enc} = 0$. This can only work for all radii r' between R_1 and R_2

if all the conductor's charge is on the inner surface to cancel the charge on the non-conducting sphere. So, the charge Q_1 on the inner surface is $Q_1 = Q_c = -2\pi\rho_0 R_1^3$, and there is no charge in the bulk of the conductor or on the outer surface:

$$\sigma_1 = \frac{Q_1}{A_1} = \frac{Q_1}{4\pi R_1^2} = -\frac{2\pi\rho_0 R_1^3}{4\pi R_1^2} = -\frac{\rho_0 R_1}{2}$$

$$\sigma_2 = 0, \quad \rho_c = 0$$

Problem 3

a. Determine the electric potential created by the semicircle at the origin.

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dQ}{R}, \quad dQ = \lambda R d\theta = \frac{Q}{\pi R} R d\theta = \frac{Q}{\pi} d\theta$$

$$\therefore V_c = \int dV = \int_0^\pi \frac{1}{4\pi\epsilon_0} \frac{Q}{\pi R} d\theta = \frac{Q}{4\pi\epsilon_0 R}$$

b. Determine the electric potential created by the rod at the origin.

$$\begin{array}{c} | \\ \hline x \quad x+dx \end{array}, \quad x < 0$$

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dQ}{(-x)}, \quad dQ = \lambda dx = \frac{Q}{R} dx$$

$$V_r = \int dV = \int_{-2R}^{-R} \frac{1}{4\pi\epsilon_0} \frac{1}{-x} \frac{Q}{R} dx$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{R} [-\ln(-x)] \Big|_{-2R}^{-R}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{R} \ln 2$$

- c. Determine the change in electric potential energy of the electron from infinity to the origin along the x -axis.

$$\begin{aligned}\Delta V &= V(0) - V(\infty) \\ &= V_c + V_r \\ &= \frac{Q}{4\pi\epsilon_0 R} + \frac{Q}{4\pi\epsilon_0 R} \ln 2 \\ &= \frac{Q}{4\pi\epsilon_0 R} (1 + \ln 2)\end{aligned}$$

$$\Delta U = (-e) \cdot \Delta V = \frac{-eQ}{4\pi\epsilon_0 R} (1 + \ln 2)$$

- d. Determine the speed of the electron as it reaches the origin, assuming it is released from rest at infinity on the x -axis.

by energy conservation , $\Delta U + K = 0$

$$\therefore \frac{1}{2} m v^2 = \frac{eQ}{4\pi\epsilon_0 R} (1 + \ln 2)$$

$$\therefore v = \sqrt{\frac{eQ}{2\pi\epsilon_0 R m} (1 + \ln 2)}$$

Problem 4

- (a) Determine the magnitude and direction of the electric field created between the two shells.

Solution. The system has cylindrical symmetry, so the electric field will point radially and its magnitude will only depend on the radial coordinate r . Choose the Gaussian surface to be a cylinder with length l between R_1 and R_2 which shares the same center axis, from Gauss's Law:

$$\oint \vec{E} \cdot d\vec{A} = E \cdot 2\pi r l = \frac{Q_{enc}}{\epsilon_0} = \frac{l\beta}{\epsilon_0}$$

Therefore we have

$$\vec{E} = \frac{\beta}{2\pi\epsilon_0 r} \hat{r}$$

And the direction is pointing radially outward.

- (b) Calculate the capacitance per unit length of the cylindrical capacitor.

Solution.

$$\Delta V = V_{in} - V_{out} = - \int_{R_2}^{R_1} \vec{E} \cdot d\vec{r} = - \int_{R_2}^{R_1} \frac{\beta}{2\pi\epsilon_0 r} dr = \frac{\beta}{2\pi\epsilon_0} \ln\left(\frac{R_2}{R_1}\right)$$

Therefore, the capacitance per unit length is given by

$$\frac{C}{l} = \frac{Q}{\Delta V} = \frac{2\pi\epsilon_0}{\ln\left(\frac{R_2}{R_1}\right)}$$

- (c) Determine the equivalent capacitance of the network of capacitors.

Solution.

$$C_2 + C_3 : C_{eq1} = \frac{C_2 C_3}{C_2 + C_3} = C/2.$$

$$(C_2 + C_3) // C_4 : C_{eq2} = C_{eq1} + C_4 = 3C/2.$$

$$[(C_2 + C_3) // C_4 + C_1] : C_{eq3} = \frac{C_{eq2} C_1}{C_{eq2} + C_1} = 3C/5.$$

- (d) Determine the energy stored in this array of capacitors if a voltage V is applied between a and b .

Solution.

$$U = \frac{Q^2}{2C} = \frac{1}{2} C_{eq3} V^2 = \frac{3}{10} C V^2$$